Brain Tumour Detection from MRI Images Using Convolutional Neural Networks and Transfer Learning

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Abstract

This report presents an evidence-based machine-learning pipeline for brain-tumour detection from magnetic resonance imaging (MRI) scans using convolutional neural networks (CNNs) and transfer learning. The task is binary image classification: each scan is labelled tumour-positive (yes) or tumour-negative (no). The dataset was obtained from Kaggle and implemented using a GitHub workflow, including a Version 1 prototype and a cleaned final notebook. The executed notebook confirms 5,450 MRI images, evenly distributed across 2,725 tumour and 2,725 non-tumour images.

The pipeline includes dataset extraction, audit, resizing, normalisation, augmentation, stratified train/validation/test splitting, baseline CNN modelling, MobileNetV2 transfer learning, metric-based evaluation and Grad-CAM explainability. MobileNetV2 outperformed the baseline CNN, achieving 93.64% accuracy, 92.20% precision, 95.35% recall, 93.75% F1-score and 0.9858 AUC, compared with the baseline CNN's 80.32% accuracy and 0.8919 AUC. The findings show that transfer learning provides stronger generalisation for this MRI classification prototype, although external validation remains necessary before clinical use.

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Abbreviations

CNN	Convolutional Neural Network
TL	Transfer Learning
MobileNetV2	MobileNet Version 2
MRI	Magnetic Resonance Imaging
Grad-CAM	Gradient-weighted Class Activation Mapping
AUC	Area Under the ROC Curve
ReLU	Rectified Linear Unit

Chapter 1: Introduction

1.1 Overview of Brain Tumour Detection from MRI Images

Brain and central nervous system cancers represent a substantial global health burden, with GLOBOCAN 2022 estimates reporting more than 321,000 new cases worldwide (Bray et al., 2024). Brain tumours also present significant diagnostic challenges because their size, location and imaging appearance vary across patients. Magnetic resonance imaging (MRI) is the primary modality for brain assessment, providing superior soft-tissue contrast without ionising radiation, and is routinely used in both initial diagnosis and post-treatment monitoring. However, expert radiological interpretation is time-consuming, subject to observer variability and increasingly constrained by specialist workforce shortages. Deep learning models can learn discriminative visual representations from labelled image archives, offering reproducible, auditable predictions that complement rather than replace clinical judgement.

This project addresses supervised binary classification. Given an MRI image, the model predicts whether the scan belongs to the tumour class (yes) or the non-tumour class (no). The significance of this task is substantial: early and accurate tumour detection directly influences treatment planning and patient survival outcomes, while inconsistent human interpretation highlights the clinical need for reproducible automated screening tools. The report follows the assessment guidelines, documenting the full pipeline rather than presenting only a final accuracy score. The objectives are to:

1. Build an auditable Kaggle-based dataset pipeline with verified class counts.
2. Implement a custom CNN benchmark as a performance baseline.
3. Compare the baseline with MobileNetV2 transfer learning (TL) to assess generalisation gains.
4. Interpret predictions using Grad-CAM explainability to support model transparency.

The repository preserves both an exploratory Version 1 notebook and a cleaned final implementation, showing progression from testing to a structured, reproducible pipeline. This objective-led structure demonstrates methodological judgement: class counts prove the dataset audit, split outputs prove experimental separation, model summaries prove architecture choices, evaluation tables prove findings, and Grad-CAM outputs support interpretability.

Chapter 2: Literature Review

2.1 CNNs, Transfer Learning and Explainability

CNNs became central to modern image classification after Krizhevsky, Sutskever and Hinton (2012) demonstrated the effectiveness of hierarchically learned convolutional features on large-scale visual recognition tasks. LeCun, Bengio and Hinton (2015) explained why: deep networks compose low-level edges and textures into abstract visual representations, directly applicable to MRI, where diagnostic signals are spatial and structural. Litjens et al. (2017) reviewed more than 300 medical image analysis studies and showed that CNN-based methods can achieve strong performance across many medical imaging tasks; however, real-world effectiveness remains critically dependent on dataset quality, validation design and clinical context factors that are frequently underreported in benchmark evaluations.

Comparing key architectures clarifies the rationale for choosing MobileNetV2. VGG-16 and VGG-19 (Simonyan and Zisserman, 2015) are well-established baselines with uniform 3×3 convolutions, but their 138 million and 144 million parameters, respectively, create substantial overfitting risk when training datasets are small or moderately sized, a characteristic of clinical imaging where annotation is expensive and time-consuming. ResNet-50 (He et al., 2016) addresses depth limitations through residual skip connections, achieving strong ImageNet performance; yet its 25 million parameters still require extensive fine-tuning data to generalise effectively to out-of-distribution domains such as medical imaging. EfficientNet (Tan and Le, 2019) achieves superior accuracy through compound scaling of depth, width and resolution, but introduces training complexity and longer convergence times that are impractical for resource-constrained portfolio experiments. None of these architectures offers an ideal balance of parameter efficiency, representational power and training speed for moderate-scale medical datasets.

MobileNetV2 (Sandler et al., 2018) was therefore selected because its inverted residuals with linear bottlenecks provide a strong balance between representational capacity and parameter efficiency. At approximately 3.4 million parameters, it is substantially smaller than ResNet-50 and VGG-16, reducing overfitting risk for a moderate-sized medical image dataset while still enabling useful pretrained visual features to be transferred from ImageNet. Explainability is a parallel requirement in medical AI: black-box predictions are clinically insufficient without supporting evidence. Grad-CAM (Selvaraju et al., 2017) generates class-discriminative spatial localisation maps from gradients flowing into the final convolutional layer, enabling visual inspection of model decisions without input perturbation. This makes it computationally efficient and well-matched to MobileNetV2, satisfying both the accuracy and interpretability requirements of this assessment.

Chapter 3: Methodology

3.1 Methodology Overview

The methodology was designed to satisfy the required machine-learning pipeline: dataset description, preprocessing, model implementation, evaluation and explainability. Numerical results and proof figures are taken from executed notebook outputs preserved in the project files and GitHub repository. The evidence strategy follows pipeline logic: each numerical claim in the methodology and results is paired with a screenshot, figure or table. The figures, therefore, function as proof of implementation, execution and evaluation rather than decoration.

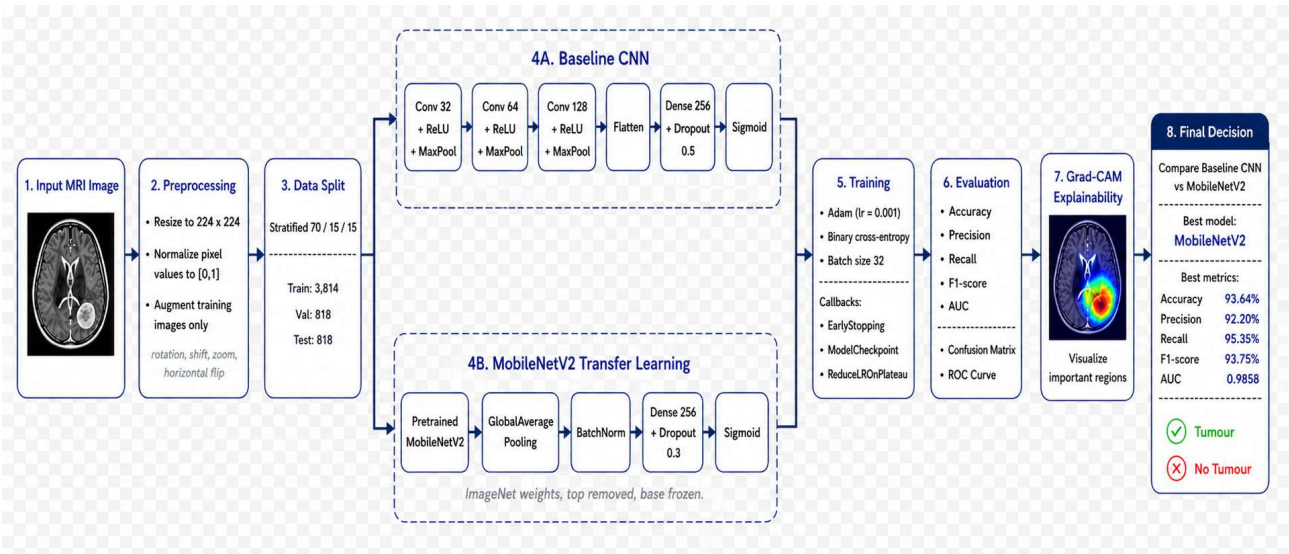

Figure 1. End-to-end algorithmic pipeline for brain tumour detection from MRI images.

The workflow shows the complete experimental process used in this study, including MRI image input, preprocessing, stratified data splitting, baseline CNN modelling, MobileNetV2 transfer learning, model training, evaluation, Grad-CAM explainability and final model selection. [See GitHub repository for executed notebook evidence.]

3.2 Dataset Description

The Brain Tumour MRI Yes or No dataset (Kaggle, n.d.), linked on the reference page as required, is an image-based binary classification dataset with two class folders: yes (tumour-positive) and no (tumour-negative). The executed GitHub notebook confirms 2,725 images per class, giving 5,450 images in total. This perfectly balanced distribution eliminates the need for oversampling or class-weighted loss functions, meaning all reported metrics directly reflect genuine discriminative ability rather than majority-class bias. The target variable is binary tumour status, and the task is supervised image classification. The dataset link is provided in the References section as required by the assessment brief.

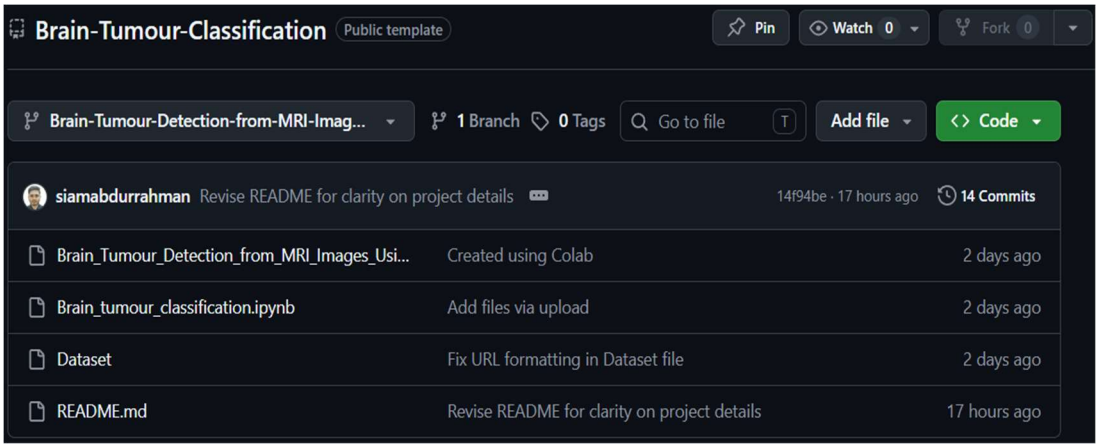

Figure 2. Repository, uploaded-notebook and dataset as evidence.

Dataset Extraction, Labels and Audit Output

```
Project folders created successfully.

Dataset extracted successfully.

no
yes

Yes images: 2725
No images: 2725
Total images: 5450

Number of unique image sizes found in sample: 23
Some example sizes: [(630, 630), (456, 538), (286, 356), (451, 539), (470, 552),
 (402, 497), (232, 217), (257, 307), (512, 512), (400, 539)]
```

Figure 3. Dataset extraction output and class-count verification from the executed notebook.

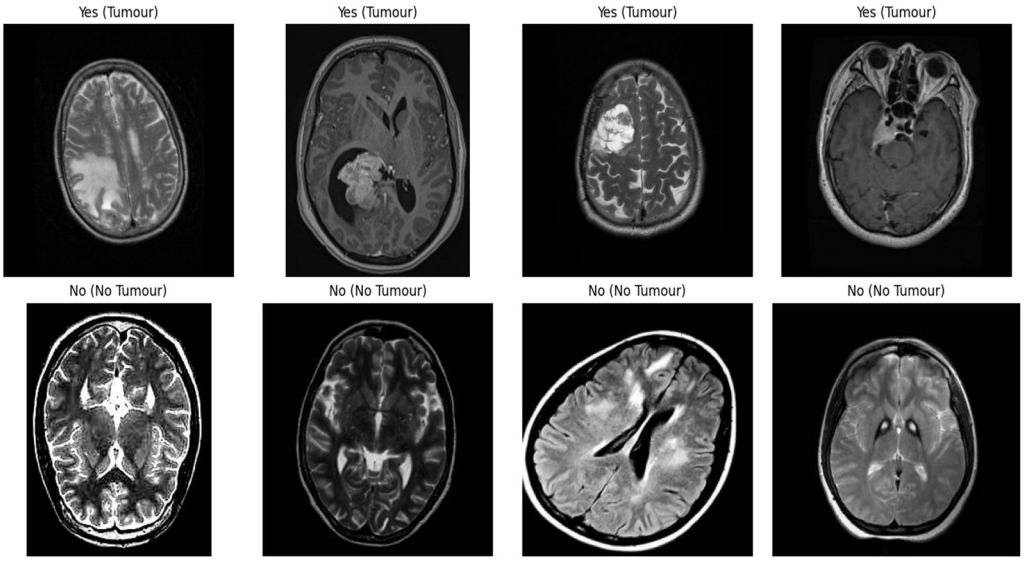

Figure 4. Representative MRI samples from the no (non-tumour) and yes (tumour) classes.

3.3 Preprocessing Pipeline

All images were resized to 224×224 pixels with three channels for compatibility with both MobileNetV2 and the custom CNN. Pixel values were normalised from [0, 255] to [0, 1] to reduce gradient magnitudes and accelerate convergence. Data augmentation, including rotation (±20°), width/height shift (0.15), zoom (0.15) and horizontal flip, was applied exclusively to the training generator using Keras ImageDataGenerator, simulating realistic imaging variation from patient positioning and scanner protocols. Validation and test generators were deliberately excluded from augmentation so that evaluation reflects performance on unmodified images, preserving the integrity of the reported metrics.

A stratified 70/15/15 split was applied independently per class, producing 3,814 training, 818 validation and 818 test images with class balance preserved across all three subsets. During Version 1 development, a folder-contamination issue was identified in which images were inadvertently assigned to multiple splits. This was resolved by fully rebuilding the split directory before training, substantially reducing the risk of test-set leakage and ensuring a clean experimental separation between development and evaluation data.

Clean Train/Validation/Test Split Proof

```
Old data_split folder deleted.
Fresh split folders created.
Original yes files: 2725
Original no files : 2725

YES split:
Train: 1907  Val: 409  Test: 409

NO split:
Train: 1907  Val: 409  Test: 409

Files copied into fresh split folders.

Final verified split counts:
train yes 1907
train no 1907
val yes 409
val no 409
test yes 409
test no 409

Train directory: /content/drive/MyDrive/brain_tumour_project/data_split/train
Validation directory: /content/drive/MyDrive/brain_tumour_project/data_split/val
Test directory: /content/drive/MyDrive/brain_tumour_project/data_split/test
Found 3814 images belonging to 2 classes.
Found 818 images belonging to 2 classes.
Found 818 images belonging to 2 classes.

Generator sample counts:
Train samples: 3814
Validation samples: 818
Test samples: 818

Class indices: {'no': 0, 'yes': 1}
Final corrected test samples: 818
```

Figure 5. Confirmed train/validation/test split after full directory rebuild, showing per-class and total counts.

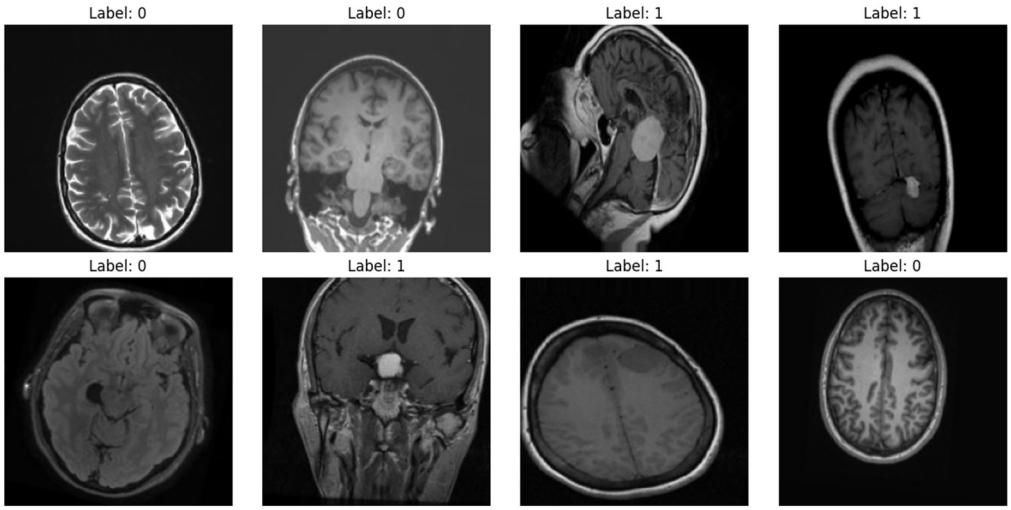

Figure 6. Augmented training examples showing rotation, shift, zoom and horizontal-flip transforms applied to training data only.

3.4 Model Implementation

Two models were implemented to enable direct performance comparison. The baseline CNN comprised three convolutional blocks (32, 64 and 128 filters; 3×3 kernel; ReLU activation; batch normalisation; 2×2 max pooling), followed by flattening, a 128-unit dense layer (ReLU), 0.5 dropout and sigmoid output. Compiled with the Adam optimiser (lr = 0.001), binary cross-entropy loss and accuracy, precision, recall and AUC metrics, the model totals 11,169,985 parameters, of which 11,169,537 are trainable. Training ran for up to 30 epochs at a batch size of 32. This large parameter count was deliberately selected to benchmark overfitting risk, motivating the transfer-learning comparison.

Baseline CNN Model Summary Proof

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
batch_normalization (BatchNormalization)	(None, 111, 111, 32)	128
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
batch_normalization_1 (BatchNormalization)	(None, 54, 54, 64)	256
conv2d_2 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 26, 26, 128)	0
batch_normalization_2 (BatchNormalization)	(None, 26, 26, 128)	512
...		
dense_1 (Dense)	(None, 1)	129

Total params: 11,169,985 (42.61 MB)

Trainable params: 11,169,537 (42.61 MB)

Non-trainable params: 448 (1.75 KB)

Figure 7. Baseline CNN model summary and parameter-count proof.

MobileNetV2 was loaded with ImageNet weights (Deng et al., 2009) with the top head excluded, and the convolutional base was frozen to preserve pretrained representations. The added classification head comprised global average pooling, batch normalisation, a 128-unit dense layer (ReLU), 0.3 dropout and sigmoid output. This configuration yields 2,427,201 total parameters with only 166,657 trainable parameters, representing a 98.5% reduction versus the baseline, substantially limiting the risk of overfitting on the moderately sized training set. Three callbacks governed training: EarlyStopping (patience = 7, monitoring validation loss), ModelCheckpoint (saving best weights) and ReduceLROnPlateau (factor = 0.3, patience = 3). These callbacks prevented overfitting and preserved the optimal model state for final evaluation.

MobileNetV2 Model Summary Proof

```
Model: "functional_27"

┏━━━
```

MobileNetV2 improved all metrics decisively. Recall rose from 70.66% to 95.35%, reducing missed tumour cases from 120 to 19 per 409 test images. The confusion matrix shows 390 true positives, 376 true negatives, 19 false negatives and 33 false positives. The AUC of 0.9858 confirms near-perfect class discrimination. Regarding fit behaviour, Figure 10 does not show a classic overfitting pattern for the baseline CNN; instead, the validation curve remains above the training curve, which is consistent with heavy augmentation and dropout making the training task harder. The safer conclusion is that the baseline underperformed the transfer-learning model, while MobileNetV2 achieved stronger and more stable test-set performance.

MobileNetV2 training and validation curves tracked closely throughout; ReduceLROnPlateau reduced the learning rate once from 0.001 to 0.0003 when validation loss stagnated, after which performance recovered. EarlyStopping terminated training before divergence. These dynamics confirm that the frozen pretrained base, combined with adaptive callbacks, effectively controlled overfitting.

MobileNetV2 Metrics and Final Comparison

```
Final MobileNetV2 training accuracy: 0.9659150242805481
Final MobileNetV2 validation accuracy: 0.9547677040100098
Final MobileNetV2 training loss: 0.09524336457252502
Final MobileNetV2 validation loss: 0.1688573956489563

MobileNetV2 Test Accuracy : 0.9364303178484108
MobileNetV2 Test Precision: 0.9219858156028369
MobileNetV2 Test Recall   : 0.9535452322738386
MobileNetV2 Test F1-score : 0.9375
MobileNetV2 Test AUC      : 0.9857604868454876

              precision    recall  f1-score   support

          no       0.95      0.92      0.94       409
         yes       0.92      0.95      0.94       409

    accuracy                           0.94       818
   macro avg       0.94      0.94      0.94       818
weighted avg       0.94      0.94      0.94       818

MobileNetV2 classification report saved at:
/content/drive/MyDrive/brain_tumour_project/results/mobilenet_classification_report.txt

[[376  33]
 [ 19 390]]

<Figure size 600x500 with 2 Axes>
      Metric  Baseline CNN  MobileNetV2  Improvement
0   Accuracy      0.803178     0.936430     0.133252
1  Precision      0.875758     0.921986     0.046228
2     Recall      0.706601     0.953545     0.246944
3   F1-score      0.782138     0.937500     0.155362
4        AUC      0.891900     0.985760     0.093860

=== Corrected Baseline CNN ===
Accuracy : 0.8031784841075794
Precision: 0.8757575757575757
Recall   : 0.706601466992665
F1-score : 0.7821380243572396
AUC      : 0.8919004549231533

=== MobileNetV2 ===
Accuracy : 0.9364303178484108
Precision: 0.9219858156028369
Recall   : 0.9535452322738386
F1-score : 0.9375
AUC      : 0.9857604868454876
```

Figure 9. MobileNetV2 classification report and final metric comparison from the executed notebook

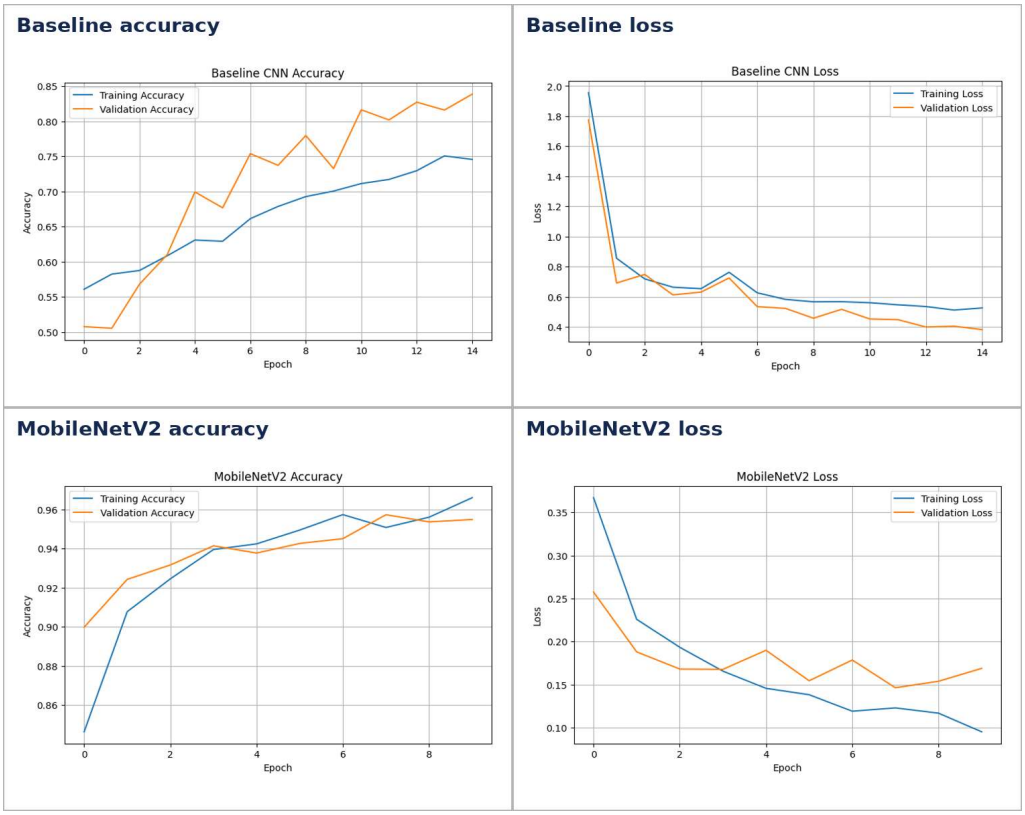

Figure 10. Training and validation accuracy/loss curves for both models.

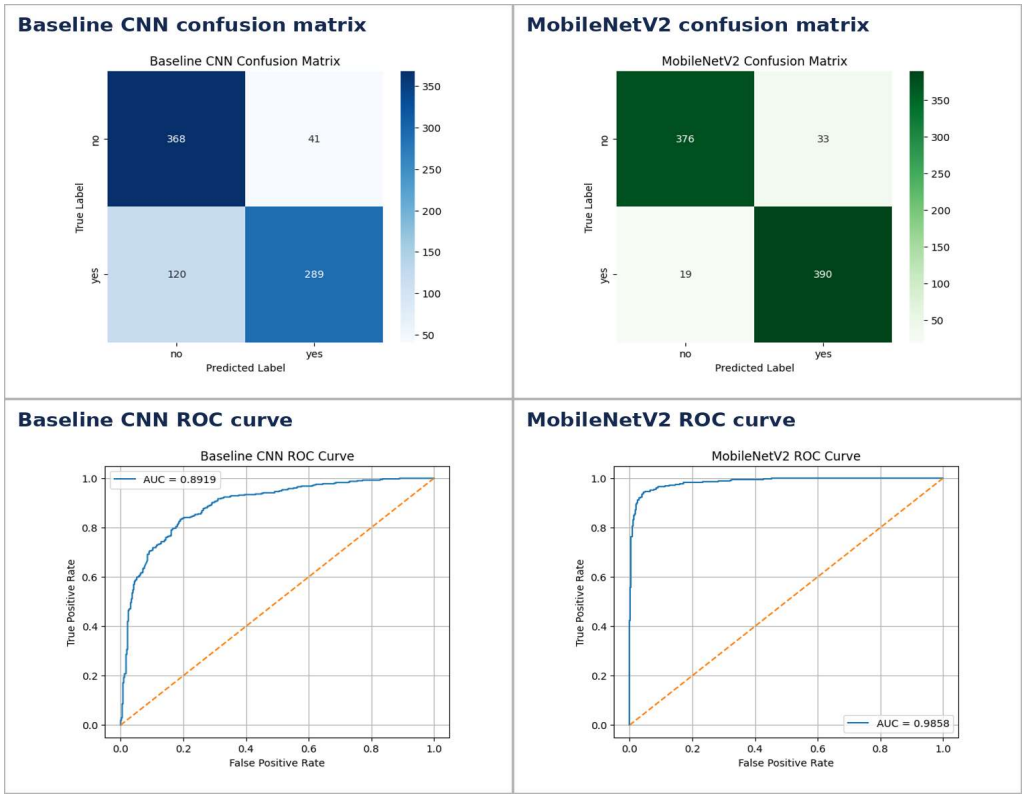

Figure 11. Confusion matrix and ROC curve proof for the baseline CNN and MobileNetV2.

3.6 Explainability

Grad-CAM (Selvaraju et al., 2017) was applied at the MobileNetV2 final convolutional layer, which the assessment brief recommends for image-based data. Grad-CAM weights spatial activation channels by their globally averaged gradients, highlighting regions that most strongly influenced each prediction class. Visualisations were generated for representative correct and incorrect predictions to support qualitative analysis of model behaviour.

True-positive examples showed concentrated heatmap activation in intracranial regions influencing the model's tumour-positive prediction, suggesting anatomically plausible localisation. True-negative examples showed diffuse, lower-magnitude activation rather than a strong focal tumour-positive signal, consistent with the absence of a salient pathological region. False-negative cases exhibited peripherally displaced activation, suggesting the model underweighted subtle tumour-boundary signals, a pattern that could guide targeted data augmentation focusing on boundary-adjacent regions. False-positive cases occasionally showed spurious activations near image borders, indicating residual sensitivity to scanner artefacts or field-of-view edges.

These findings suggest the model frequently attends to anatomically plausible regions; however, Grad-CAM provides qualitative evidence rather than quantitative localisation proof. Future work should compare Grad-CAM activation overlap against radiologist-annotated bounding boxes using intersection-over-union (IoU) metrics to assess spatial accuracy rigorously. Nevertheless, the error patterns identified through Grad-CAM analysis offer actionable guidance for data curation and model refinement.

Grad-CAM Selection and Error Proof

```
Using Grad-CAM target layer: out_relu

Number of corrected test filepaths: 818
Number of corrected test labels: 818

Correct tumour cases: 390
Correct non-tumour cases: 376
Missed tumour cases (false negatives): 19
False positives: 33

Selected indices for Grad-CAM: [409, 410, 0, 1, 441]

Baseline false negatives: 120
MobileNetV2 false negatives: 19
MobileNetV2 recall: 0.9535452322738386
Grad-CAM file: /content/drive/MyDrive/brain_tumour_project/figures/mobilenet_gra
dcam_explainer.png
```

Figure 12. Grad-CAM target-layer, test-file and selected-case proof from the executed notebook.

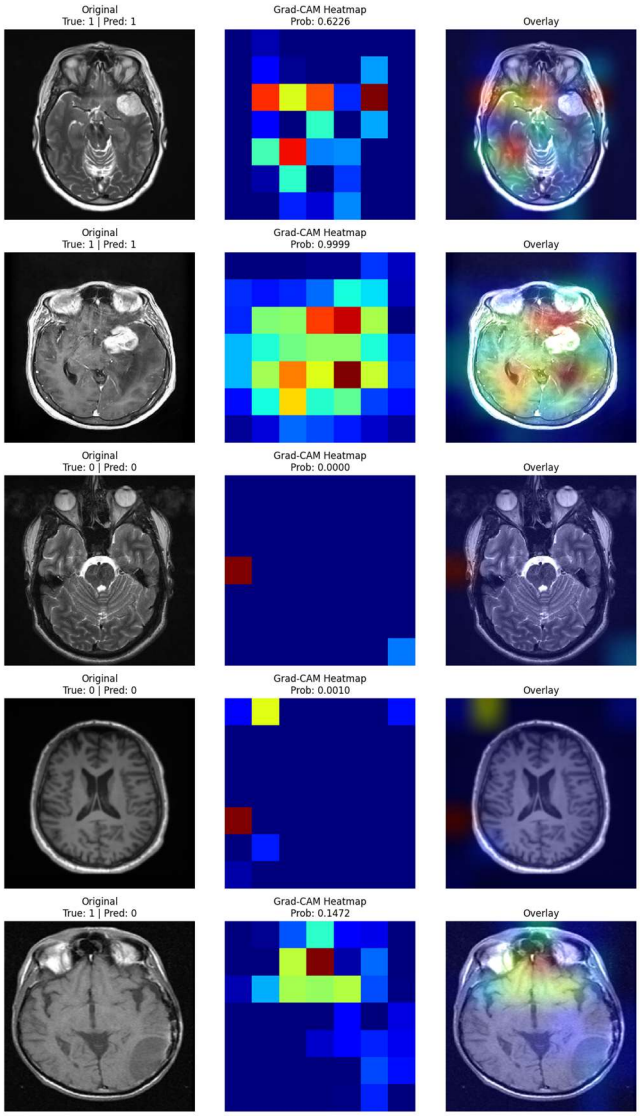

Figure 13. Grad-CAM overlay grid showing original MRI, heatmap and composite for representative correct and incorrect predictions.

Chapter 4: Discussion and Future Directions

4.1 Critical Discussion, Limitations and Future Improvements

The results confirm that transfer learning is substantially more effective than a custom CNN trained from scratch for this dataset. MobileNetV2 achieved stronger recall (+24.69 pp), F1-score (+15.54 pp) and AUC (+0.0939) while using 98.5% fewer trainable parameters during feature-extraction training. The scale of the recall improvement is particularly noteworthy: the baseline CNN missed approximately 120 tumour-positive cases per 409 test images compared with 19 for MobileNetV2, a clinically significant difference in a screening context. This gap can be attributed to the limited capacity of a randomly initialised network to learn effective tumour discriminators

from approximately 3,800 training images, a task for which the pretrained MobileNetV2 base, having learned general visual feature hierarchies from ImageNet, is far better positioned. The pipeline is methodologically stronger for including class auditing, a contamination-corrected split, augmentation, multi-metric comparison, learning curve analysis and Grad-CAM evidence.

However, the work remains a prototype with several important limitations. The public dataset may not represent the diversity of real clinical MRI scanners, acquisition protocols, tumour types or patient populations encountered in practice. The binary label also simplifies clinical reality significantly, as tumours differ by grade, location, subtype and imaging characteristics in ways that a single yes/no label cannot capture. No patient-level splitting was performed, meaning near-duplicate scans from the same patient may appear in both training and test sets, potentially inflating reported metrics.

Future improvements should include: patient-level splitting and exact or near-duplicate checking before experimental separation; external hospital dataset validation to assess domain transferability; calibration analysis to ensure that model confidence scores are meaningful; segmentation-based localisation with IoU validation against expert annotations; and systematic comparison with EfficientNet, ResNet-50 and DenseNet architectures. Ethical deployment in a clinical environment would require radiologist oversight at all decision points, patient privacy safeguards, robust dataset governance, prospective clinical validation and calibrated probability reporting so that borderline cases can be flagged for human review rather than forced into a binary decision.

Chapter 5: Conclusion

5.1 Summary and Closing Remarks

This report developed an evidence-based brain tumour classification pipeline using MRI images, CNN modelling and MobileNetV2 transfer learning. The notebook evidence demonstrates dataset extraction, balanced class counts, contamination-corrected train/validation/test splitting, preprocessing with augmentation, model construction, comprehensive multi-metric evaluation and Grad-CAM explainability. MobileNetV2 achieved the strongest performance across all metrics, with 93.64% accuracy, 95.35% recall, 93.75% F1-score and 0.9858 AUC, representing substantial improvements over the custom CNN baseline. The project demonstrates that transfer learning can significantly improve MRI-based tumour classification compared with training from scratch on a moderately sized dataset. Nevertheless, the system should be treated as a reproducible academic prototype rather than a clinically deployable diagnostic tool until externally validated on patient-level clinical data and assessed under the ethical and regulatory standards required for medical AI deployment.

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Code availability:

The full implementation notebook and supporting files are available at:
https://github.com/siamabdurrahman/Brain-Tumour-Classification

The repository contains both the exploratory Version 1 prototype and the cleaned final implementation, allowing the development process and final structured pipeline to be reviewed.